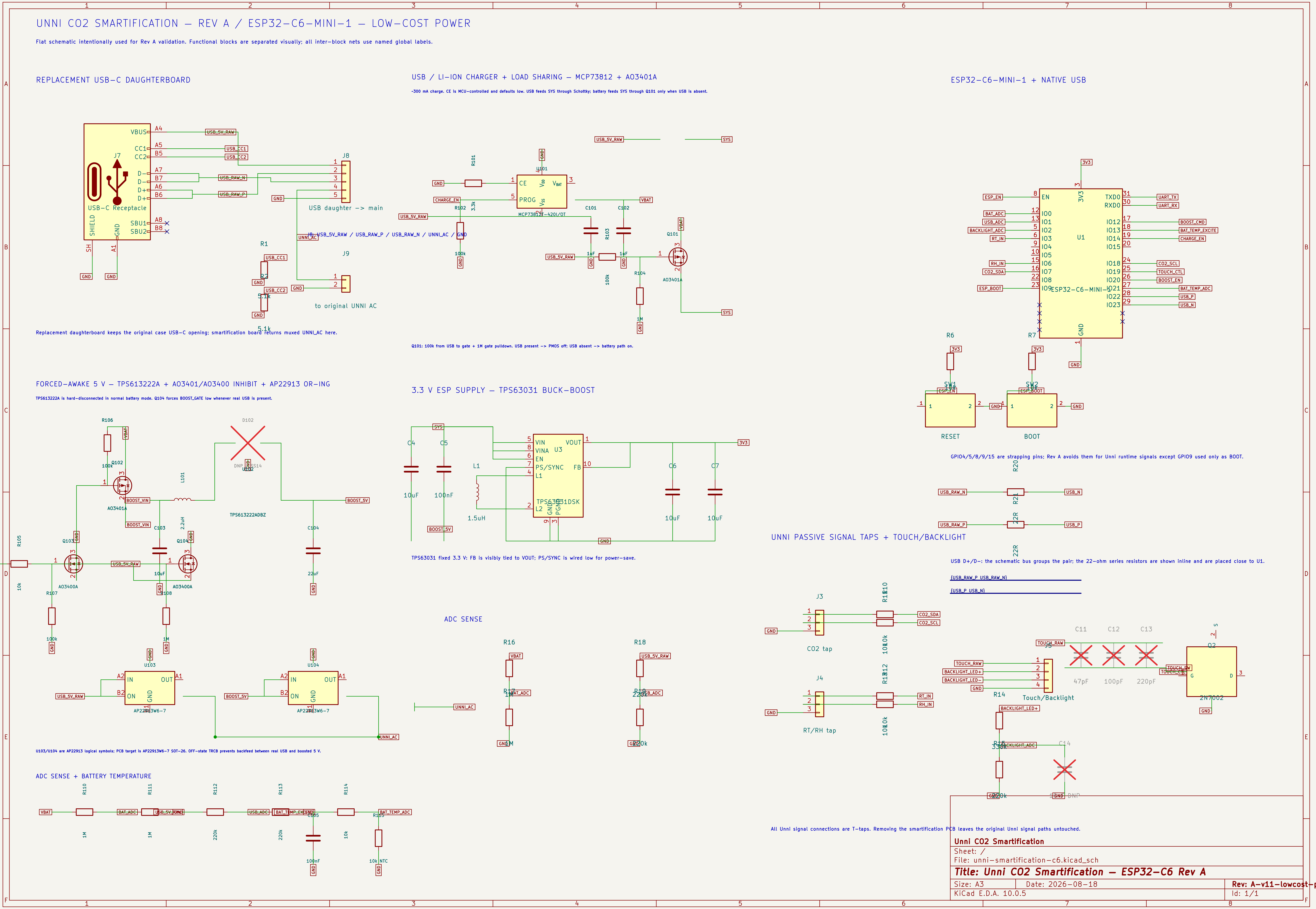

UNNI CO2 SMARTIFICATION – REV A / ESP32-C6-MINI-1 – LOW-COST POWER
Flat schematic intentionally used for Rev A validation. Functional blocks are separated visually; all inter-block nets use named global labels.
REPLACEMENT USB-C DAUGHTERBOARD
USB-C Receptacle
USB daughter -> main
J8: USB_5V_RAW / USB_RAW_P / USB_RAW_N / UNNI_AC / GND
to original UNNI AC
Replacement daughterboard keeps the original case USB-C opening; smartification board returns muxed UNNI_AC here.
USB / LI-ION CHARGER + LOAD SHARING – MCP73812 + AO3401A
~300 mA charge. CE is MCU-controlled and defaults low. USB feeds SYS through Schottky; battery feeds SYS through Q101 only when USB is absent.
MCP73812T-420I/OT
Q101: 100k from USB to gate + 1M gate pulldown. USB present -> PMOS off; USB absent -> battery path on.
ESP32-C6-MINI-1 + NATIVE USB
ESP32-C6-MINI-1
RESET
BOOT
GPIO4/5/8/9/15 are strapping pins; Rev A avoids them for Unni runtime signals except GPIO9 used only as BOOT.
FORCED-AWAKE 5 V – TPS613222A + AO3401/AO3400 INHIBIT + AP22913 OR-ING
TPS613222A is hard-disconnected in normal battery mode. Q104 forces BOOST_GATE low whenever real USB is present.
TPS613222ADBZ
U103/U104 are AP22913 logical symbols; PCB target is AP22913W6-7 SOT-26. OFF-state TRCB prevents backfeed between real USB and boosted 5 V.
3.3 V ESP SUPPLY – TPS63031 BUCK-BOOST
TPS63031DSK
TPS63031 fixed 3.3 V: FB is visibly tied to VOUT; PS/SYNC is wired low for power-save.
ADC SENSE
ADC SENSE + BATTERY TEMPERATURE
UNNI PASSIVE SIGNAL TAPS + TOUCH/BACKLIGHT
USB D+/D-: the schematic bus groups the pair; the 22-ohm series resistors are shown inline and are placed close to U1.
{USB_RAW_P USB_RAW_N}
{USB_P USB_N}
CO2 tap
RT/RH tap
Touch/Backlight
All Unni signal connections are T-taps. Removing the smartification PCB leaves the original Unni signal paths untouched.
Unni CO2 Smartification
Sheet: /
File: unni-smartification-c6.kicad_sch
Title: Unni CO2 Smartification – ESP32-C6 Rev A
Size: A3
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